LLM Linguistic Evolution: Language and Cooperation in LLM Dyads

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2026-05-02

Abstract

Can natural-language exchange between LLM agents become behaviourally load-bearing, or does it merely accompany decisions already fixed by model and prompt? We introduce a controlled dyadic paradigm pairing a role-swapping repeated Trust Game with an iterated myth-writing task. Across GPT-5-Nano and Claude Sonnet 4.5, with variation in task order and action-channel noise, the implicit-channel design produces only small and conditional behavioural effects: across 24 evaluable cells, myth-writing lifts cooperation in a few upward-noise cells, destabilises cooperation under reciprocity-flipping (bootstrap) noise, and is null in most cells. Linguistic dynamics are nonetheless robust independent of behaviour: between-agent embedding cosine converges in 18 of 21 cells, and lag-1 cross-agent cooperativity correlations are positive in every cell. A 2x3 follow-up factorial (visibility x content direction) on the bootstrap-noise destabilisation isolates the active mechanism: making the partner's myth visible in the game prompt reverses the destabilisation (mean shift from -7.7 to +1.5; std 11.0 to 3.5), with content semantics (cooperative, neutral, adversarial) explaining less than 5% of the variance. The intervention is regime-conditional, producing no effect at no-noise baseline. The reading is consistent with a Chwe-style common-knowledge anchor and inconsistent with both a competing-prior account and a content-as-cooperation-cue account. Inter-agent language is load-bearing on cooperation, but through visibility, not semantics.

1 1. Introduction

LLM agents increasingly interact with other LLMs through natural language: assistants negotiate, agent-society simulations stage repeated encounters, and model-versus-model harnesses test social behaviour [Park et al., 2023; Zhou et al., 2023]. Yet most LLM-agent studies still treat the exchanged text as scaffolding while measuring numeric outcomes such as cooperation rates, donations, or payoffs [Akata et al., 2023; Lorè and Heydari, 2023]. The agents talk, but the talk is rarely the object of analysis. We ask whether language in cooperating LLM dyads is *load-bearing*: does it change cooperation, or merely accompany it?

The motivating analogy is human narrative. Stories, myths, and norms can align expectations and stabilise cooperation; importantly, they may reduce variance in cooperative behaviour rather than raise mean cooperation (§2.3). We do not assume LLMs share the human mechanism. The analogy instead motivates an experimental probe: keep the linguistic artefact separable, let it evolve across repeated interaction, and ask whether it predicts or changes game behaviour.

We pair two same-base-model LLM agents in a role-swapping repeated Trust Game [Berg et al., 1995], optionally interleaved with an iterated myth-writing task. We vary task order and action-channel noise, with a small neutral-framing pilot for sanity checking (§A.3), and measure both game behaviour and linguistic content. Figure 1 summarises the design. The core contrast is deliberately simple: compared with a game-only dyad, does adding an evolving myth channel raise cooperation, consolidate a strategy the dyad has already found, or remain linguistically active but behaviourally inert?

Our contribution is a controlled paradigm for crossing iterated linguistic interaction with repeated economic games, plus a preliminary empirical map of where the paradigm has behavioural headroom. The current evidence supports a cautious conclusion. Model identity and payoff-channel perturbations dominate mean cooperation. Myth-writing does not behave like a reliable cooperation switch. It sometimes shifts payoffs, but the sign and magnitude depend on the model and channel condition; its more plausible signal is consolidation and linguistic convergence within dyads. The central claim is therefore narrow: inter-agent language is not mere decoration, but this first design does not yet show a robust myth-induced cooperation lift. A controlled visibility-by-content factorial on the bootstrap-noise destabilisation isolates the active mechanism: making the partner's myth visible in the game prompt reverses the harm, while content semantics (cooperative, neutral, or adversarial) account for less than 5% of the variance, consistent with a Chwe-style common-knowledge anchor (§4.6).

2 2. Background

2.1 2.1 LLM agents in cooperation games

A growing experimental literature examines LLM agents in canonical economic games [Aher et al., 2022; Akata et al., 2023; Fontana et al., 2024; Horton, 2023; Leng and Yuan, 2023; Lorè and Heydari, 2023]. LLMs often cooperate in single-shot prisoner's dilemmas and trust games, but with strong model-, prompt-, and task-dependence; they show distinctive failure modes in pure coordination tasks; and framing manipulations — persona, social context, elicited identity — reliably move behaviour [Serapio-García et al., 2023].

Against this backdrop, Tewolde et al. [Tewolde et al., 2026] complicate the picture: stronger reasoning models consistently defect in single-shot dilemmas, but contractual side-payments and third-party mediation raise cooperation while repetition and reputation do not; we test a communicative channel Tewolde did not examine: storytelling.

A smaller strand moves past single-shot snapshots to repeated and multi-agent settings, asking whether cooperation is sustained when defection pressure accumulates. Piatti et al.'s commons-game study is representative: LLM populations in a shared-resource setting are evaluated on whether they avoid collapse [Piatti et al., 2024]. The Diplomacy / CICERO line is the high-water mark for natural-language-mediated cooperation in AI, showing that language-using agents can negotiate, form alliances, and sustain coordination at human level in a demanding mixed-motive game [Bakhtin et al., 2022]. But language in CICERO is engineered toward task performance; the messages are a means, not an object of study.

2.2 2.2 Cultural evolution and iterated learning in artificial agents

Human laboratory studies show that cumulative cultural transmission, imposed under a learnability bottleneck, drives communicative systems toward compositional structure and compression rather than mere convergence [Kirby et al., 2008, 2015; Raviv et al., 2019; Tamariz and Kirby, 2016]. We inherit two features of the paradigm: round-by-round transmission with adaptation pressure, and systematic study of the transmitted artefact across generations.

A recent agenda extends this view to LLM populations as substrates where transmission–drift–selection dynamics can be studied on artefacts humans recognise as cultural [Brinkmann et al., 2023; Perez et al., 2024]. Three strands bear on our design: transmission-chain experiments with LLMs reproduce known human content biases, showing the substrate carries culturally relevant structure [Acerbi and Stubbersfield, 2023]; multi-generational LLM populations exhibit cultural-evolution-like dynamics [Perez et al., 2024]; and Vallinder & Hughes extend the paradigm to cooperation itself, showing cooperative dispositions can undergo cultural evolution across generations of LLM agents in a donor game [Vallinder and Hughes, 2024].

2.3 2.3 Narrative, myth, and cooperation in humans

What cultural-evolution work on LLMs has not yet addressed is the specific linguistic form hypothesised to carry cooperative content in humans: narrative. Storytelling correlates with — and plausibly supports — cooperation in small-scale human societies [Smith et al., 2017]. Wiessner's study of Ju/'hoan (!Kung) conversation found that daytime talk is dominated by economic coordination and sanctioning gossip, but roughly 80% of nighttime conversation consists of stories — of known people, distant kin, and cultural institutions — suggesting that storytelling is the register in which norms, extended-network trust, and shared imagination are specifically rehearsed [Wiessner, 2014]. Cultural-evolution and adaptationist accounts treat shared narratives as functional units supporting cooperation rather than incidental by-products of language: compressed carriers of behavioural prescriptions and coordination devices that align expectations and stabilise reciprocity [Bietti et al., 2018; Boyd et al., 2011; Henrich, 2016; Scalise Sugiyama, 2001].

Several causal mechanisms have been proposed for how narrative supports cooperation. Two are central to our probe: compression of normative content into transmissible, memorable units [Boyd et al., 2011; Henrich, 2016]; and coordination via common knowledge, where publicly shared stories generate the higher-order beliefs needed for equilibrium selection [Chwe, 2001]. Others — notably social simulation [Mar and Oatley, 2008] and credibility-enhancing displays [Henrich, 2009] — pick out distinct causal pathways but fall outside our design.

Myths are short, narrative, value-laden, and explicitly transmissible — they compress a moral stance into a unit a partner can recognise, internalise, and rewrite. This makes them an unusually clean probe for the content a partner takes from your previous turn, since the same unit simultaneously targets both the compression and common-knowledge mechanisms while admitting iterative adaptation across rounds. We make no claim about social simulation or CREDs, which would require designs observing behaviour-beyond-text or separating private from public exchange (see §3.3).

The human evidence is correlational and small-N, and faces a reverse-causation worry: cooperative groups may simply have the slack for elaborate storytelling. We do not claim LLMs reproduce the same mechanism; the human regularity motivates the probe, and the LLM design lets us manipulate the linguistic channel directly — sidestepping (but not resolving) the human-side identification problem.

3 3. Methods

3.1 3.1 Overview

Each run pairs two agents that share a base model, temperature, persona, and memory budget, with a fixed task order, game-parameter preset, and seed. Our headline question is whether adding the myth-writing task changes either the *level* of cooperation in the trust game (a cooperation-lift hypothesis) or its *across-seed dispersion* (a strategy-consolidation hypothesis); we therefore report median reward deltas and variance changes for myth-present cells against matched game-only controls. Configurations expand from `config/experiments_noisy.yaml` and runs save under `data/json/noise_experiments/`.

3.2 3.2 Repeated trust game

We use a role-swapping repeated Trust Game with endowment $E = \$5$ and multiplier $m = 3$. Each game round has two sequential moves: the investor chooses $s_t \in [0, E]$ to send to the trustee; the trustee receives $3s_t$ and chooses $r_t \in [0, 3s_t]$ to return. Per-round payoffs are

$$\pi_t^I = E - s_t + r_t, \qquad \pi_t^T = 3s_t - r_t.$$

Roles alternate every game round, so both agents experience both sides. Unless otherwise noted, runs use ten game rounds, temperature 0.8, a neutral persona, and memory capacity three. The memory buffer is shared across game and myth turns, so interleaving a myth task reduces the number of recent game exchanges still visible in context. On the first game round the investor sees only the rules and endowment; on later rounds each agent is shown the previous round's sent, received, and returned amounts, its payoff, and current cumulative balance.

3.3 3.3 Myth-writing task

In myth-present conditions, each round also contains a creative-writing turn. On the first myth round, both agents write a 200-word myth on the unconstrained topic `anything`; on later rounds, each agent sees its own

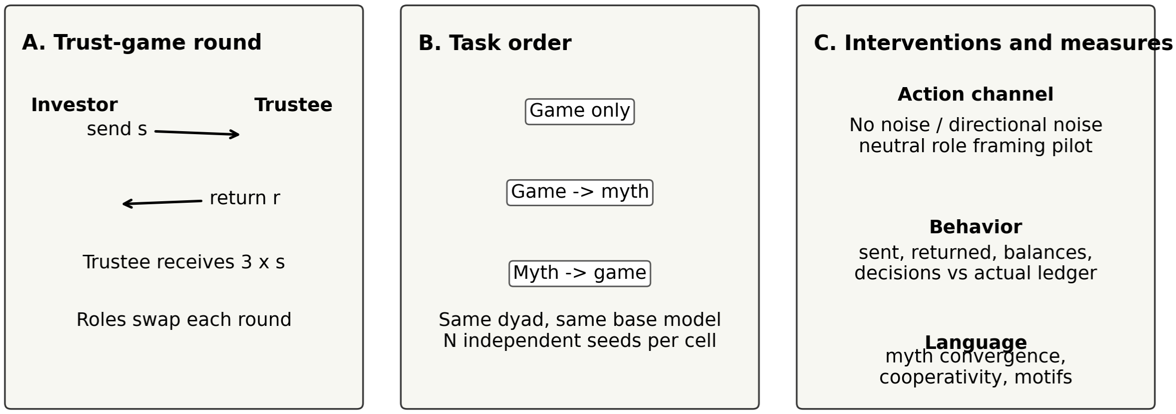

Figure 1: Experimental design. Two same-model agents alternate investor and trustee roles in a repeated trust game. Runs either contain only the game, interleave game before myth-writing, or interleave myth-writing before the game. The action channel can be unperturbed, directionally noised, or neutrally reframed; outputs include both game ledgers and myth text.

previous myth and the partner's previous myth and is instructed to adapt them in its own way. This creates a dyadic iterated-transmission channel separate from the payoff-bearing game. The game prompts never quote the partner's myth directly: cross-task influence flows only through the shared message buffer, where the agent's own prior myth and the partner myth it saw during writing remain in recent context when later game decisions are made.

3.4 3.4 Conditions

The main draft uses two direct-provider models: OpenAI `gpt-5-nano` and Anthropic `claude-sonnet-4.5`. Gemini pilot outputs are excluded from the main analysis as the run is incomplete.

Task order has three game-containing levels: game-only, game then myth, and myth then game. The main no-noise baseline is `baseline_v4_mem3_direct`, with 15 seeds per model and task order. A neutral-framing pilot, `neutral_framing_v4_pilot`, replaces investor/trustee wording with ROLE A/ROLE B allocation wording, with three seeds per cell.

Noise is an environmental perturbation on the actual transfer ledger: the model chooses a decision amount, the environment optionally perturbs it, and the perturbed value is what arrives, enters payoffs, updates balances, and appears in subsequent prompts. Unperturbed choices are logged separately as `sent_decision` and `returned_decision` (see §A.1 for full ledger bookkeeping). We use three action-channel interventions. Directional uniform noise adds $\epsilon \sim \mathcal{U}(0, k)$ or $\epsilon \sim \mathcal{U}(-k, 0)$ to both send and return actions and clamps to range, with $k = 5$ for the completed manipulation-check runs. Bootstrap noise replaces the returned amount with the maximum possible return every round, an artificial reciprocity-support diagnostic. A deterministic-max variant (every-round application of the maximum perturbation) was run as a pilot but is under-replicated and excluded from the main analysis (§A.4). Informed variants append a system-prompt note flagging that transfers may be perturbed and that earnings reflect arrived amounts.

Two further channel-design variants probe the cross-task mechanism (results in §4.6). The **partner-myth injection** variant quotes the partner's most recent myth verbatim inside the trust-game prompt, opening a direct cross-task channel that the implicit design (§3.3) keeps closed. The **forced-reasoning** variant adds a system-prompt addendum requiring 2–3 sentences of reasoning before the JSON action, used only with GPT-5-Nano to elicit the reasoning prose Claude emits by default. Both variants run on all four bootstrap noise conditions; partner-myth injection is also crossed with three myth-content directions (neutral, cooperative, adversarial) for the §4.6 factorial.

3.5 3.5 Measures and analysis

Behavioural measures are computed from raw JSON states: final dyad reward after ten game rounds, average sent and returned amounts (both arrived and decision), return ratios, and per-cell dispersion across seeds. The primary myth contrast is the median final-dyad-reward difference between each myth-present task order and the matched game-only cell within the same model and condition, with non-parametric bootstrap percentile intervals over seeds. Each cell is then labelled by a joint mean-shift × variance-ratio rule. The Δmean 95% CI is read as *positive*, *negative*, or *null* (strictly above, strictly below, or spanning zero); the myth-to-game variance-ratio 95% CI is read as *consolidation*, *destabilization*, or *null* relative to one. The two axes combine into *lift+consolidation*, *lift*, *consolidation*, *null*, *harmful* (negative mean, null variance), and *destabilizing* (any other combination involving negative mean or destabilization). Cells with $n_{\mathrm{myth}} < 10$ are flagged *missing* and excluded.

Linguistic measures span lightweight and embedding-based proxies, all reproducible without API calls: cooperative-lexicon density, coarse pronoun ratios, lag-1 cross-agent cooperativity correlations, and dictionary-based coinage detection over the myth chain. We additionally compute between-agent embedding cosine similarity per round using `sentence-transformers/all-mpnet-base-v2` (normalized) and a lexical proxy for myth content appearing in game-response prose: the share of game-response tokens matching the agent's own preceding myth vocabulary. LLM-judge similarity is deferred to follow-up analysis. All manuscript tables and figures are generated from raw final JSON files (see §A.2).

4 4. Results

4.1 4.1 Cross-model behavioural baselines (no-noise)

The two models occupy distinguishable but no longer dramatically different no-noise regimes. In the v4 direct-provider configuration with $T = 10$, mem-3, and the unconstrained `anything` myth topic, GPT-5-Nano reaches a mean cumulative dyad balance of 69.2 in the game-only cell (median 71.0, std 4.6, $N = 15$), with `game_myth` and `myth_game` cells slightly higher (71.1 and 72.3). Claude-Sonnet-4.5's matched no-noise cell is under-replicated ($N = 2$); the prior `baseline/` corpus gives a mean of 55.7 (median 55.0, std 2.6, $N = 15$). Maximum possible dyad reward is 150. This revises an earlier working assumption: pilot work with GPT-5-Nano via OpenRouter showed apparent floor-locking at mutual defection, but with the direct provider the same model reaches roughly half the dyad ceiling without any noise intervention. The cross-model gap is therefore moderate, around 14 points, not the chasm the pilot suggested. We retain Claude as the higher-cooperation regime and GPT-5-Nano as the more variable one, but both exhibit substantive cooperation at baseline [Serapio-García et al., 2023]. Across-seed dispersion drops monotonically across task orders even at baseline (std 4.6 to 1.1 for GPT-5-Nano), so the variance-reduction pattern in §4.3 is not specific to the noise interventions.

4.2 4.2 Effect of noise on the cooperation profile

Direction of noise dominates the noise effect. Across the four implemented mechanisms, positive (uniform +5) noise pushes both models near the cooperation ceiling: Claude reaches a game-only mean of 73.5 (informed: 74.2), GPT-5-Nano 72.5 (informed: 72.4). Negative (uniform −5) noise depresses Claude to 29.3 (informed: 30.4) but lifts GPT-5-Nano relative to its baseline, to 37.9 (informed: 38.4); agents perceive losses as smaller than they are, sustaining trust longer.

Bootstrap noise is qualitatively distinct. It replaces the trustee's returned amount with the maximum possible return every round, masking actual reciprocation. GPT-5-Nano's game-only mean under bootstrap is 66.6 (informed: 67.9), well above no-noise. Bootstrap is therefore not just an "upward" variant; it removes the trustee-return signal entirely, a property that becomes load-bearing in §4.3. Informedness has not produced a large qualitative shift: agents told the channel is noisy still respond mainly to the realised ledger, and informed and uninformed cells differ by 1–2 points across most cells. We treat it as a robustness check. Gemini cells from the prior pilot are out of scope per §3.4 and left for the EMNLP follow-up.

4.3 4.3 Effect of myth-writing on game behaviour (the headline question)

The myth effect is conditional on the noise regime, not universal. Of 24 evaluable (model × noise × myth-order) cells in v4, the joint Δmean and variance-ratio classification (with bootstrap CIs) gives 4 lift+consolidation, 1 pure consolidation, 2 pure lift, 13 null, 2 harmful, and 2 destabilizing — seven positive cells, four negative, thirteen null. Lift+consolidation lives entirely in upward-noise cells; destabilization lives entirely in GPT-5-Nano × bootstrap. Figure 2 summarises the full classification.

The strongest positive cells (lift+consolidation, with both mean and variance CIs strict) sit entirely in upward-noise conditions, and their variance reductions run between 3× and 20× relative to the matched game-only cell. The negative cells, by contrast, are all GPT-5-Nano × bootstrap; the four harmful or destabilizing cells share Δmean estimates roughly between −7 and −10 points of dyad reward. Per-cell estimates and bootstrap CIs are tabulated in Table 1 (§A.4).

The asymmetry is structural. Under upward noise, agents observe a perceived signal at least as positive as the underlying decision, and the dyad locks onto a coordination point [Boyd et al., 2011]. Under bootstrap, the perceived signal contradicts the underlying interaction and myth introduces a competing structured prior. All eight negative-noise cells are null.

4.4 4.4 Linguistic dynamics in the myth chain

Between-agent embedding cosine convergence is robust across cells. Pairwise cosine between Agent_1 and Agent_2 myths at the same round (sentence-transformers/all-mpnet-base-v2, normalised) shows a positive slope in 18 of 21 cells, strongest in GPT-5-Nano × negative_5 × `myth_game` (slope +0.027/round [0.016, 0.038], cosine 0.42 → 0.74 across 10 rounds), GPT-5-Nano × bootstrap × `myth_game` (+0.023/round), and Claude × positive (informed) × `myth_game` (+0.021/round). Convergence holds even where the game outcome is null or harmful, so linguistic alignment is not contingent on behavioural improvement [Fontana et al., 2024].

Lag-1 cross-agent cooperativity correlations are positive in every cell. Pearson r between Agent_A's cooperativity-lexicon score in round-t myth and Agent_B's in round $t+1$ ranges 0.27–0.58 in mean per-dyad max-direction, with 27–73% of dyads above $|r| > 0.5$. The strongest cell is GPT-5-Nano × positive (informed) × `myth_game` (lag1_max mean 0.58 [0.43, 0.71], 73% of dyads above 0.5); the pilot's Claude r ≈ 0.72 sits at the upper tail of a clearly non-zero distribution and is representative, not anomalous. Both convergence and lag-1 transmission are summarised across cells in Figure 4.

Coinages persist within myth chains but do not transfer into game-reasoning text in any cell; the pilot observation of cross-task coinage leakage does not replicate at corpus scale.

4.5 4.5 Coupling between game behaviour and myth content

Claude emits extensive game-response prose threaded with own-myth vocabulary: 78–82% of round-level responses contain at least one content word from the preceding myth (mean 5–7 unique overlaps per response), and 100% of runs have at least one such reference, with theme-lexicon hits (story, spirit, elder, sacred, ancestor, ritual) reaching 10–33% of responses. GPT-5-Nano, by default, emits no reasoning prose at all — its `game_responses[ag].content` is the bare JSON action (e.g. `{"send": 3}`, ~12 characters), which makes the metric methodologically undecidable in that configuration. An A3 forced-reasoning addendum closes the gap: GPT-5-Nano under A3 matches Claude's any-hit rate at 80–82% (Figure 5).

Three layers of evidence point the same direction: between-agent embeddings systematically converge in 18/21 cells; cooperativity-lexicon scores transmit lag-1 between agents in every cell; and for Claude, myth-vocabulary content is present in essentially every game response. The carriers differ across semantic embedding, narrow lexical scoring, and thematic vocabulary, but all three indicate that the myth channel is doing measurable work [Leng and Yuan, 2023; Piatti et al., 2024]. The §4.3 mean-and-variance picture is consistent with this: language is doing real work, but its sign depends on whether it converges with or competes against the noise channel [Smith et al., 2017].

Figure 2: 3x3 classification (mean shift x variance shift, bootstrap CIs) for every (model x noise x myth-order) cell. Green = positive myth effect, red = negative, gray = null/missing. Lift+consolidation lives in upward-noise cells; destabilization in GPT-5-Nano x bootstrap.

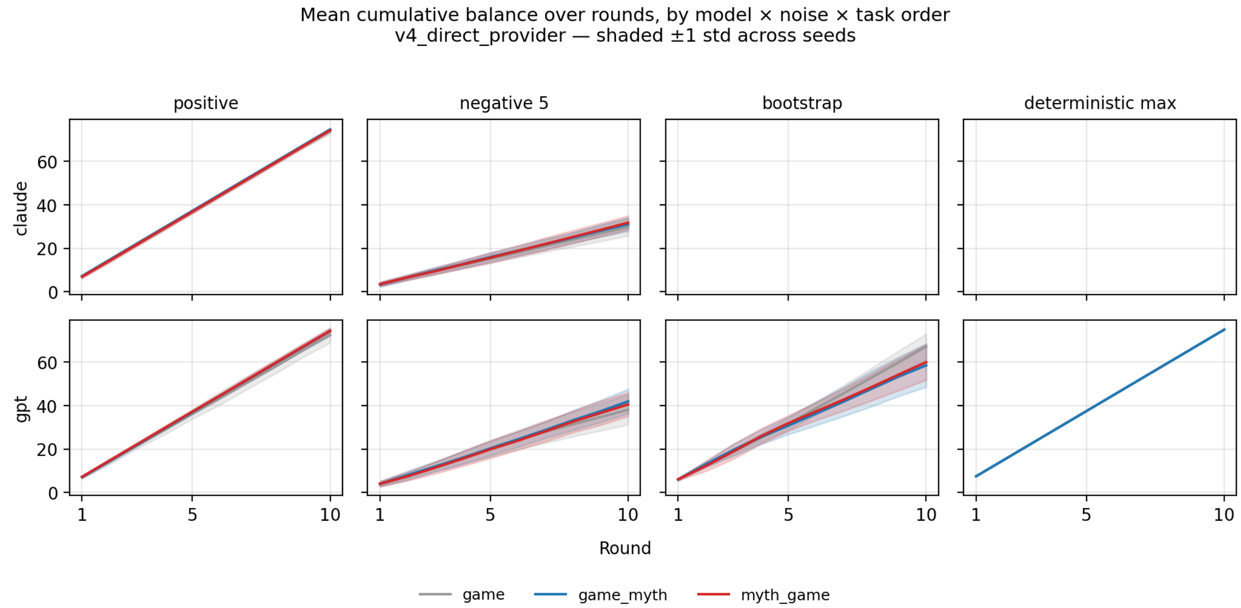

Figure 3: Mean cumulative balance over rounds, faceted by (model x noise) with one line per task order. Shaded ribbons are +/-1 std across seeds. The bootstrap-noise harm pattern is visible in the bottom row: gray (game-only) sits above blue/red (myth-present).

4.6 4.6 Tightening the cross-task channel — a 2x3 mechanism factorial

We opened the implicit channel of §3.3 by quoting the partner's most recent myth verbatim inside the trust-game prompt — a **partner-myth injection** variant — and ran it on the four noise conditions of §4.3 (all GPT-5-Nano).

In the bootstrap cells where myth was harmful, partner-myth injection reverses the destabilisation. Bootstrap $\times$ `game_myth`: baseline anything-myth Δmean -7.7 against a game-only mean of 58.9 (std 11.0); with injection, Δmean +1.5 (mean 68.1, std 3.5). The pattern holds in `myth_game` (Δmean -7.6 $\rightarrow$ +1.8; std 9.2 $\rightarrow$ 3.6). Across-seed variance collapses roughly threefold and mean cooperation returns to baseline. In the positive-noise cells where myth was already lifting, injection adds no detectable extra effect (Δmean change $<$ 0.2 across all four positive cells). The intervention is regime-conditional: it acts only where the implicit channel was failing.

To discriminate visibility from content, a follow-up factorial varied the myth topic across three directions: anything-myth (neutral content), cooperative-themed (`reciprocity_oath`, a story about honouring reciprocal obligations), and adversarial (`trickster_exploitation`, a story about being exploited by a defector). Each crossed with the visibility manipulation (no injection vs injection), yielding [N_CELLS] cells of fresh data, all GPT-5-Nano $\times$ bootstrap, $N = 15$ per cell.

Visibility dominates content direction. The 2×3 factorial on bootstrap $\times$ `game_myth` (uninformed):

Content / visibility	No injection	+ Injection
Anything (neutral)	58.9 $\pm$ 11.0 (harmful)	**68.1** $\pm$ **3.5** (rescue)
Cooperative	57.4 $\pm$ 12.4 (harmful)	**69.8** $\pm$ **2.9** (rescue)
Adversarial (defection-themed)	n/a	**66.6** $\pm$ **5.5** (rescue)

(The adversarial $\times$ no-injection cell was not run; the no-injection harm pattern was already established in the neutral and cooperative arms.) All three injection cells land in the rescue zone (66–70); both no-injection

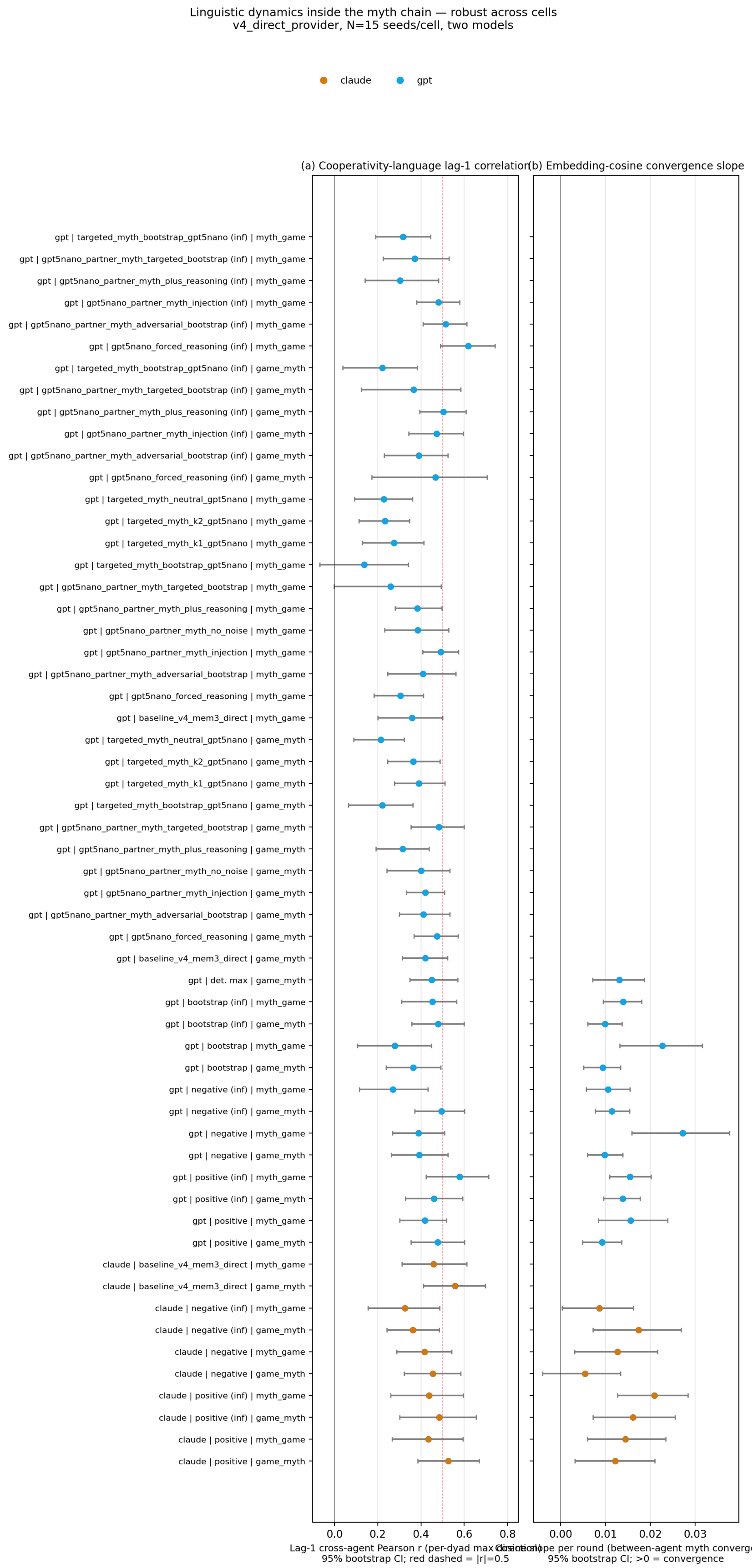

Figure 4: Lag-1 cross-agent cooperativity correlation (left) and embedding-cosine convergence slope (right) across all (model x noise x myth-order) cells, with Claude and GPT-5-Nano overlaid.

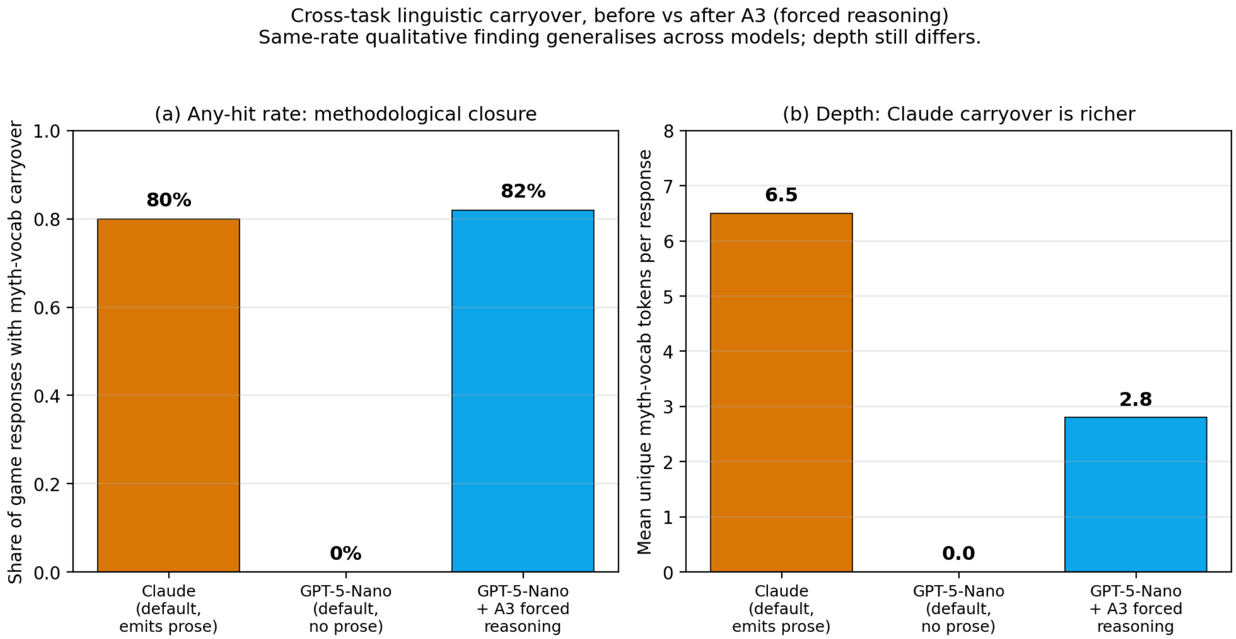

Figure 5: Any-hit rate (left) and depth (right) of own-myth-vocabulary carryover into game responses, comparing Claude default, GPT-5-Nano default, and GPT-5-Nano under A3 forced reasoning.

cells stay in the harmful zone (~58). The same pattern recurs in `myth_game` and in the informed bootstrap variants. Cooperative content modulates the rescue marginally upward (~+2 over neutral) and adversarial content marginally downward (~-2); content direction explains less than 5% of variance, visibility the rest (Figure 6).

Two further controls anchor the mechanism. An A1 × no-noise baseline produces no effect (Δmean -0.8 to +0.9 across two cells, $N = 30$ runs, 0 failures), ruling out the alternative that injection simply lifts cooperation by adding more partner-related text; if that were the mechanism, no-noise cells would also lift. An A1+A3 combination (injection plus a system-prompt addendum requiring 2–3 sentences of reasoning before each JSON action) erases A1's rescue in the bootstrap cells (Δmean returns to ~-13 against A1). Inspection shows GPT-5-Nano with forced reasoning surfaces explicit cost–benefit calculation rather than coordination; "net zero last round → send less" patterns dominate. The Claude reasoning-prose carryover from §4.5 may therefore be partly correlational: cooperative reasoning correlates with cooperative behaviour without necessarily causing it [Vallinder and Hughes, 2024].

We read the mechanism as follows. Under bootstrap, agents see a perceived signal that contradicts the underlying game; with the partner's myth visible, both have common-knowledge access to a shared narrative anchor and resolve the contradiction by coordinating on it [Chwe, 2001]. Cooperative content provides a modest amplifier and adversarial content does not subtract enough to break the rescue, so content is a small modulator and visibility carries the work; the competing-prior interpretation is falsified by both content arms (cooperative and adversarial), since visibility rescues regardless of direction.

5 5. Discussion

5.1 5.1 What we found and what we didn't

The corpus yields three layered findings. Cross-model variation in baseline cooperation is moderate, around 14 points of dyad reward, but its direction depends on the noise regime: in the matched no-noise corpus GPT-5-Nano is somewhat above Claude, under positive noise both models reach the cooperation ceiling, and under negative noise GPT-5-Nano sustains higher cooperation than Claude (§4.1–4.2). The myth-writing

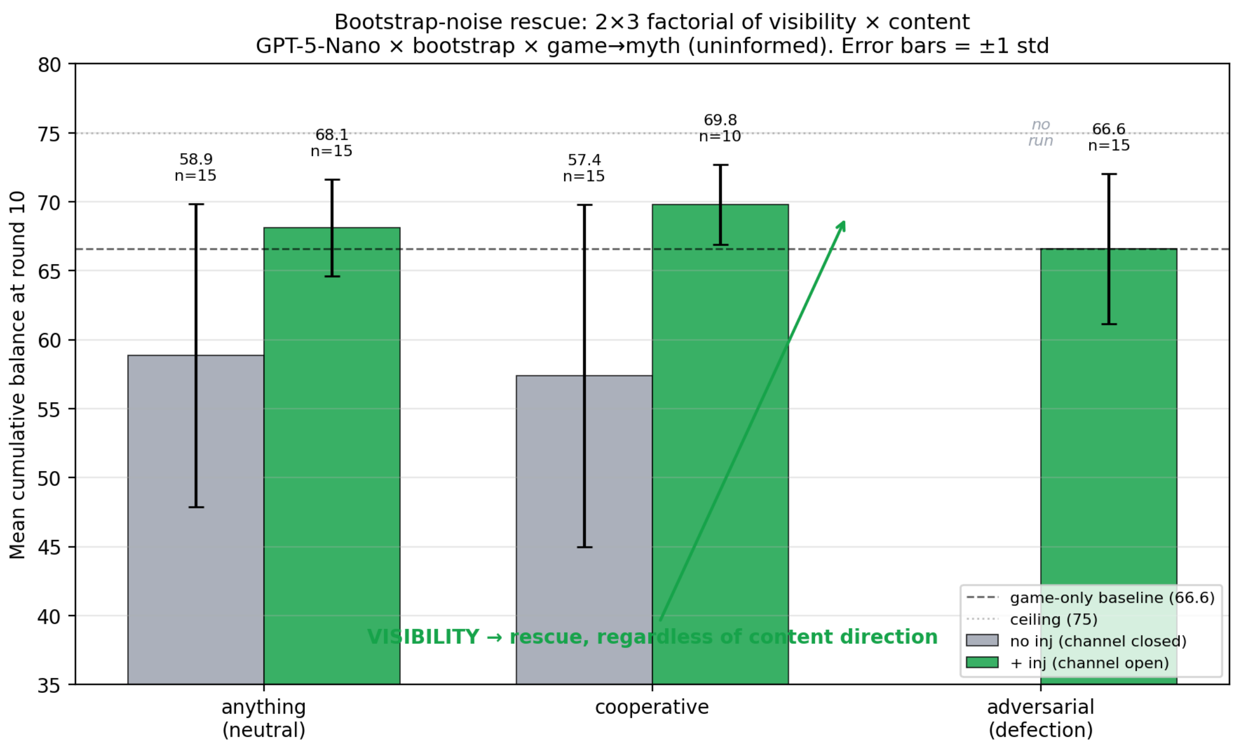

Figure 6: Bootstrap-noise rescue under partner-myth visibility for GPT-5-Nano x bootstrap x game_myth (uninformed): visibility produces the rescue across all three content directions, with content direction modulating the rescue only marginally.

effect on game payoffs is heterogeneous and conditional on noise: of 24 evaluable (model × noise × myth-order) cells, seven are positive, four are negative, and thirteen are null (§4.3). Linguistic dynamics inside the myth chain are robust regardless of behavioural outcome: between-agent embedding cosine converges in 18 of 21 cells, lag-1 cross-agent cooperativity correlations are positive in every cell, and Claude's game-response prose threads its own myth vocabulary into roughly four-fifths of rounds (§4.4–4.5).

About half of cells show no detectable myth effect on mean cumulative reward, and we report this as the central empirical fact rather than burying it. The §1 question was whether language between LLM agents is *load-bearing* on cooperation; the data answer "yes, but conditionally and asymmetrically." Where the noise regime gives the dyad headroom and the perceived signal is useful, myth-writing produces a small mean lift and a 3× to 20× drop in across-seed dispersion in the strongest cells. Where the noise regime systematically contradicts the action signal, it actively harms cooperation. The narrow form of the §1 thesis survives; the broader form, a uniform cooperation lift, is rejected.

5.2 5.2 The active mechanism is common-knowledge visibility, not content semantics

The §4.3 mean-and-variance pattern admits multiple mechanism candidates, and the §4.6 factorial decomposes between them. Three candidates motivated the follow-up: myth as a *competing structured prior* whose narrative content competes with the noise channel; myth as a *cooperative-content cue* whose semantics directly guide cooperation; and myth as a *common-knowledge anchor* in the sense of Chwe [Chwe, 2001], where both agents seeing the same narrative creates a shared reference that resolves the noise channel's contradictions. Under a visibility × content factorial, the first predicts that more visible myth makes things worse, the second that cooperative content rescues whether or not it is shared, and the third that visibility rescues regardless of content direction.

The data support the third candidate and falsify the first two. Visibility dominates content direction: common-knowledge injection rescues bootstrap cells whether content is neutral, cooperative, or adversarial, with content modulating an order of magnitude less (cell means and stds in §4.6 and Figure 6). Two controls anchor the reading. The same injection produces no detectable effect at no-noise baseline, ruling out a generic-text-lift alternative; the intervention is regime-conditional. Adding forced reasoning prose on top of injection (A1+A3) erases the rescue, with GPT-5-Nano surfacing cost–benefit reasoning that depresses cooperation below the no-injection baseline. This complicates the §4.5 prose-carries-content reading and suggests Claude's higher cooperation may be partly substrate (Claude-as-cooperator) rather than carryover.

Stated as a falsifiable claim: under bootstrap-style noise, agents observe a perceived reciprocation signal that systematically contradicts their own game-math; with both agents' previous narrative writing made common-knowledge in the game prompt, both have access to the same shared reference and resolve the contradiction by coordinating on it. The shared reference is the active ingredient; its semantic direction is a small modulator. This is consistent with the cultural-evolution literature on narrative as a coordination device [Boyd et al., 2011; Chwe, 2001]: narratives stabilise cooperation by being publicly shared, not primarily by being prescriptively cooperative.

5.3 5.3 Cross-model variation, narrowed honestly

The submitted scope is a two-model claim: the v4 direct-provider corpus contains Claude and GPT-5-Nano, with Gemini cells deferred to a re-run under the corrected noise pipeline. The two models are distinguishable in mean cooperation (~14 points of dyad reward at no-noise, §4.1) but both reach near-ceiling cooperation under positive noise. Cross-model variation in social-game behaviour is a recurring finding [Fontana et al., 2024; Leng and Yuan, 2023; Serapio-García et al., 2023]; our contribution is more specific in that variation persists *under matched experimental conditions* and shows up not only in mean cooperation but in the very different shapes of the linguistic channel. Claude emits extensive game-response prose threaded with own-myth vocabulary; GPT-5-Nano emits only the structured action with no surrounding prose at all (§4.5). Pooling across frontier models would have hidden both the bootstrap-noise harm pattern (GPT-5-Nano-only) and the reasoning-prose presence (Claude-only); future LLM-game work should report findings with model-stratified statistics.

5.4 5.4 Cross-task influence: language is in the loop, measurably

The §4.3–4.5 corpus deliberately does not inject the partner's myth into the game prompt (§3.3); the implicit channel is thin and easily overshadowed. Three layers of evidence nevertheless show it doing measurable work: between-agent embedding convergence (§4.4), positive lag-1 cross-agent cooperativity correlations (§4.4), and Claude's myth-vocabulary carryover into game-response prose (§4.5). Coinages from myths do not reappear in game `reason` text (§4.4); carryover travels through thematic vocabulary rather than coined neologisms (§4.5). Even with the implicit channel deliberately constrained (§3.3), three convergent signals leak through; the §4.6 visibility result clarifies that the dominant pathway is common-knowledge access rather than the implicit channel itself.

5.5 5.5 Limitations

The horizon is short ($T = 10$), well below typical iterated-learning paradigms. The unit is a same-model two-agent dyad with no population dynamics, partner-switching, or cross-model pairing, and all v4 cells use the neutral persona and the unconstrained `anything` topic. The noise interventions are artefacts designed to manufacture headroom rather than mimic naturally-occurring noise. Our linguistic measures are conservative and do not pick out *what* is converging. GPT-5-Nano emits no reasoning prose in this configuration, so the §4.5 cross-task channel is observable only for models that emit reasoning text alongside the action. We have no human baseline; comparisons run LLM dyads against other LLM dyads, so claims about model-personality magnitudes or about narrative-as-coordination-device would benefit from human anchors.

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6 Appendix

6.1 A.1 Simulation flow and v4 noise bookkeeping

A batch run resolves one `experiment_set` from `config/experiments_noisy.yaml` into the cartesian product of model, prompt template, persona, task order, myth topic, game-parameter preset, and seed. Each cell is one independent dyad. The current direct-provider runs are launched with `LLM_PROVIDER=direct` through `scripts/run_noisy_missing.py`, which skips final JSONs that already exist and reruns only missing outputs.

Within each run, `TrustGameNoisy` assigns roles by round, fixes the move order to investor then trustee, and appends a per-round record to `conversation_history`. When the investor responds, the JSON `send` decision is extracted as `sent_decision`. If noise applies to sends, the environment adds the configured perturbation and clamps to the valid range; this actual amount is stored as `sent`, determines the trustee's received amount, and is shown to the trustee. When the trustee responds, the same process is applied to the `return` decision, yielding `returned_decision` and `returned`.

Payoffs and cumulative balances are computed from `sent` and `returned`, not from the unperturbed decisions. Later prompts also summarize the actual previous ledger. Thus the v4 state has one behavioural ledger plus decision fields, rather than parallel actual/communicated ledgers. A noisy round records `sent_noise`, `returned_noise`, the raw unclipped noise draws, final balances, full structured model responses, and any myth text.

If a game response cannot be parsed, the run aborts and writes an error checkpoint. Myth-writing calls receive one retry. The analysis excludes `.results.json`, `.checkpoint.json`, and `.error.json` files from final tables, while writing discovered error files to `analysis/draft_artifacts/error_manifest.csv`.

6.2 A.2 Analysis artifacts

The draft artifacts are generated by:

```
MPLCONFIGDIR=/tmp/nlet-matplotlib XDG_CACHE_HOME=/tmp/nlet-cache \
PYTHONPYCACHEPREFIX=/tmp/nlet-pycache \
python3 scripts/build_neurips_draft_artifacts.py
```

The outputs are:

- `manifest.csv`: one row per final run JSON included in the analysis.
- `error_manifest.csv`: discovered error checkpoint files.
- `behavioral_summary.csv`: per model x condition x task-order behavioural summaries.
- `myth_effects.csv`: bootstrap myth-present deltas against matched game-only controls.
- `linguistic_rounds.csv` and `linguistic_summary.csv`: lexical convergence and cooperativity proxies.
- `draft_findings.md`: compact regenerated findings summary for paper-writing.

6.3 A.3 Noise diagnostics

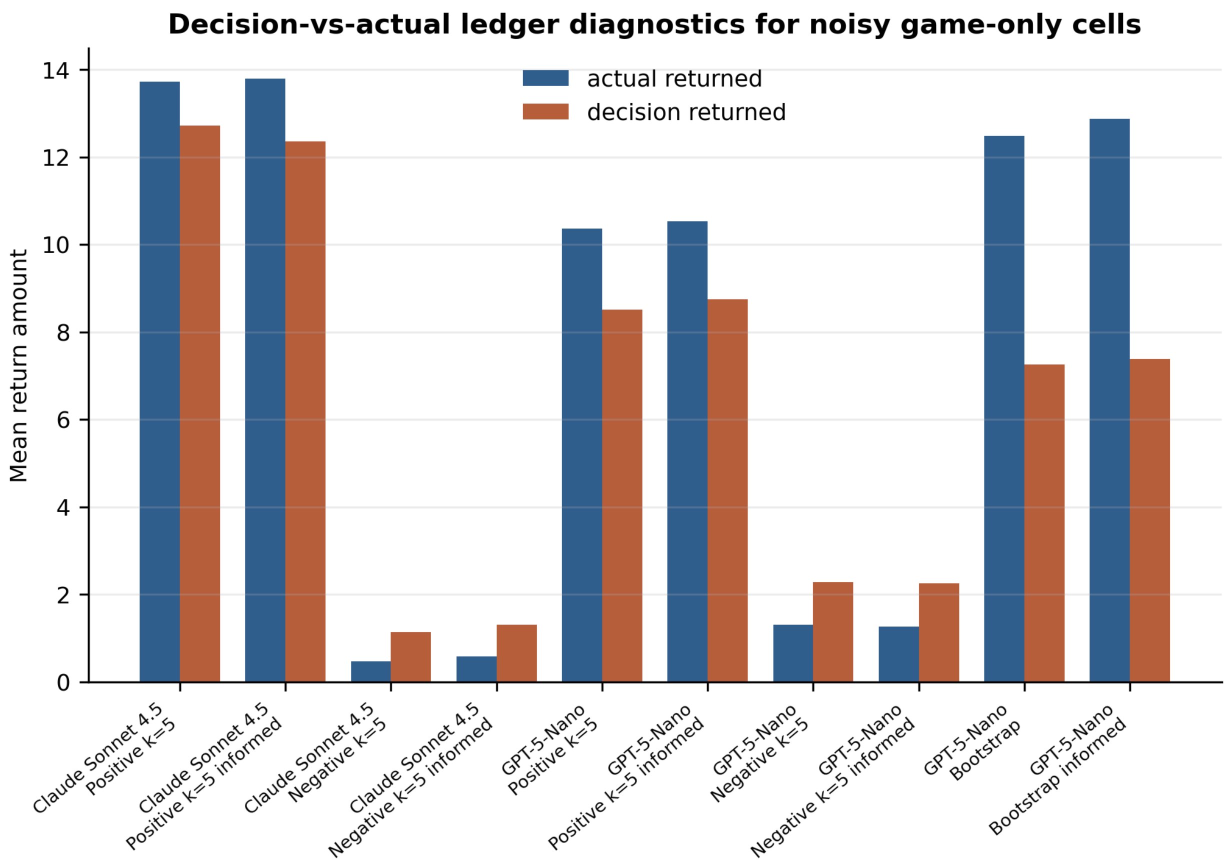

Figure 7: Decision-vs-actual returned amounts in noisy game-only cells. The separation is largest when environmental perturbations or bootstrap returns move the ledger away from the model's requested return.

6.4 A.4 Per-cell delta summary (main 24-cell analysis)

Table A.1 summarises the per-cell mean shift and variance-ratio estimates for the 24 evaluable (model x noise x myth-order) cells of the main §4.3 analysis: 4 baseline (no-noise) cells plus 20 noise cells, excluding one GPT-5-Nano deterministic-max cell ($n_{\mathrm{myth}} = 7$, classified as missing). Δmean is the difference between the matched myth-present cell and its game-only baseline; var ratio is the across-seed variance ratio (myth/game).

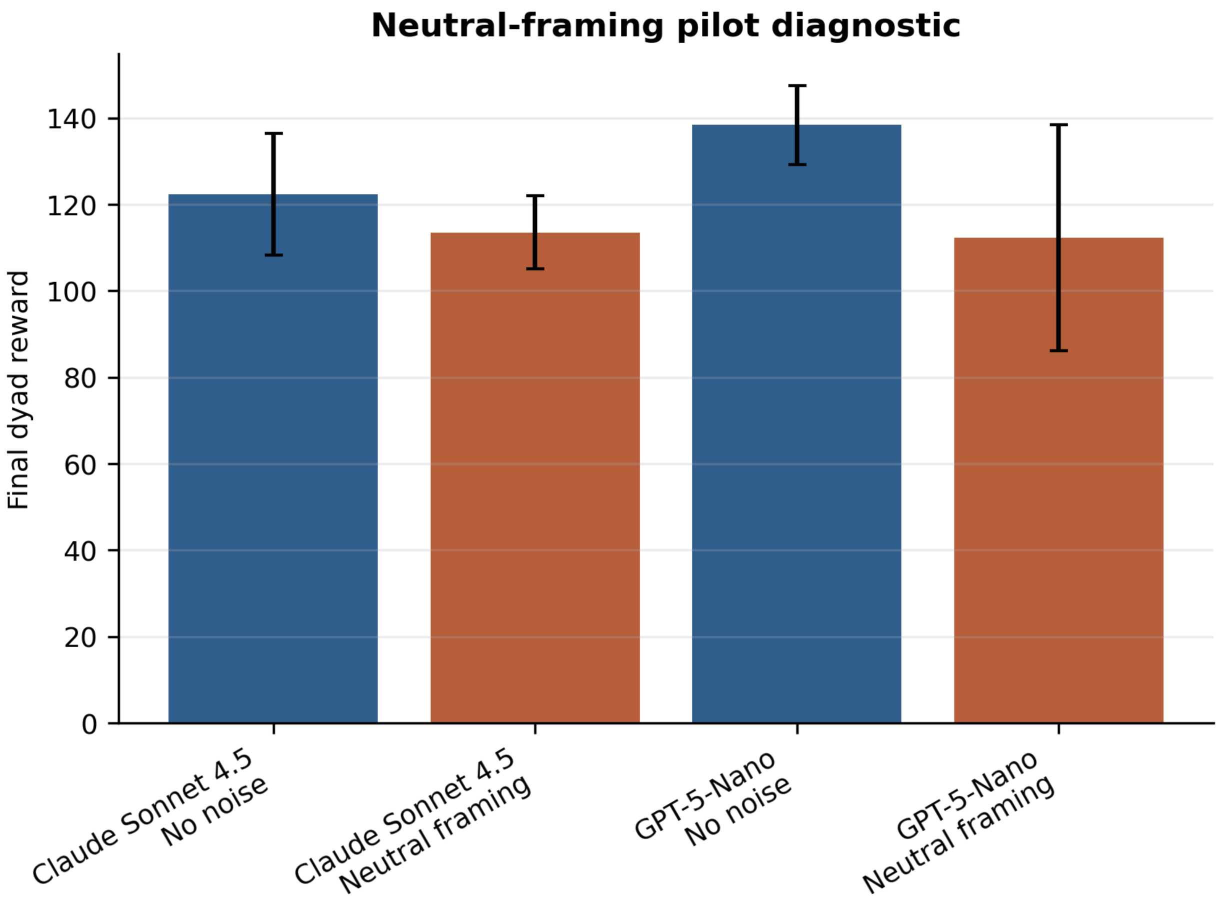

Figure 8: Neutral-framing sanity check: behavioural distributions under the neutral prompt are equivalent to those under the standard prompt across matched cells.

95% bootstrap CIs use 2000 resamples. Classifications follow the joint mean-shift x variance-ratio rule described in §3.5.

Table 1: Per-cell Δmean and variance-ratio estimates (with 95% bootstrap CIs) for the main §4.3 analysis.

Model	Noise	Task order	Δmean (95% CI)	Var ratio (95% CI)	Class
claude	no-noise	game->myth	-2.13 [-6.34, +2.00]	0.52 [0.16, 1.01]	null
claude	no-noise	myth->game	-0.93 [-5.14, +3.53]	0.62 [0.33, 1.10]	null
claude	positive	game->myth	+1.11 [+0.29, +1.96]	0.27 [0.02, 1.76]	lift
claude	positive	myth->game	+0.80 [-0.13, +1.75]	0.49 [0.09, 3.30]	null
claude	positive (inf)	game->myth	+0.59 [+0.21, +0.96]	0.31 [0.03, 0.66]	lift+consolidation
claude	positive (inf)	myth->game	+0.15 [-0.33, +0.61]	1.06 [0.36, 2.13]	null
claude	negative_5	game->myth	+1.29 [-1.60, +4.16]	0.79 [0.19, 2.66]	null
claude	negative_5	myth->game	+2.48 [-0.36, +5.08]	0.61 [0.22, 2.07]	null
claude	negative_5 (inf)	game->myth	+1.18 [-1.26, +3.33]	0.36 [0.13, 1.62]	null
claude	negative_5 (inf)	myth->game	+1.51 [-0.98, +3.93]	0.67 [0.21, 3.00]	null
gpt5-nano	no-noise	game->myth	+1.87 [-0.60, +4.33]	0.36 [0.06, 1.78]	null
gpt5-nano	no-noise	myth->game	+3.07 [+0.87, +5.33]	0.06 [0.02, 0.23]	lift+consolidation
gpt5-nano	positive	game->myth	+1.65 [-0.20, +3.97]	0.07 [0.01, 0.65]	consolidation
gpt5-nano	positive	myth->game	+2.08 [+0.26, +4.52]	0.05 [0.00, 0.50]	lift+consolidation
gpt5-nano	positive (inf)	game->myth	+2.23 [+0.99, +3.57]	0.08 [0.01, 0.25]	lift+consolidation
gpt5-nano	positive (inf)	myth->game	+1.86 [+0.37, +3.41]	0.50 [0.02, 2.19]	lift
gpt5-nano	negative_5	game->myth	+4.35 [-0.56, +9.00]	0.39 [0.10, 1.09]	null
gpt5-nano	negative_5	myth->game	+2.20 [-2.80, +7.19]	0.48 [0.18, 1.46]	null
gpt5-nano	negative_5 (inf)	game->myth	+3.37 [-0.65, +7.20]	1.41 [0.62, 3.58]	null
gpt5-nano	negative_5 (inf)	myth->game	+2.70 [-0.81, +6.23]	0.98 [0.41, 2.73]	null
gpt5-nano	bootstrap	game->myth	-7.73 [-14.07, -1.13]	2.09 [0.72, 15.95]	harmful
gpt5-nano	bootstrap	myth->game	-7.60 [-13.33, -1.73]	1.46 [0.49, 10.76]	harmful
gpt5-nano	bootstrap (inf)	game->myth	-9.67 [-14.60, -4.53]	7.50 [3.62, 16.21]	destabilizing

Model	Noise	Task order	Δmean (95% CI)	Var ratio (95% CI)	Class
gpt5-nano	bootstrap (inf)	myth->game	-6.87 [-11.13, -2.93]	4.76 [1.45, 11.18]	destabilizing

6.5 A.5 Breadth of partner-myth injection (A1) across cells

The A1 effect is regime-conditional rather than uniform. Across all eight GPT-5-Nano noise cells, A1 produces no detectable behavioural shift in the positive or negative_5 cells (Δmean within ± 0.9), and only the bootstrap cells show a substantial Δmean of approximately +9.3 (Figure 9). This pattern situates the rescue described in §4.6 as a property of the bootstrap regime specifically, not a generic lift from added partner-related text.

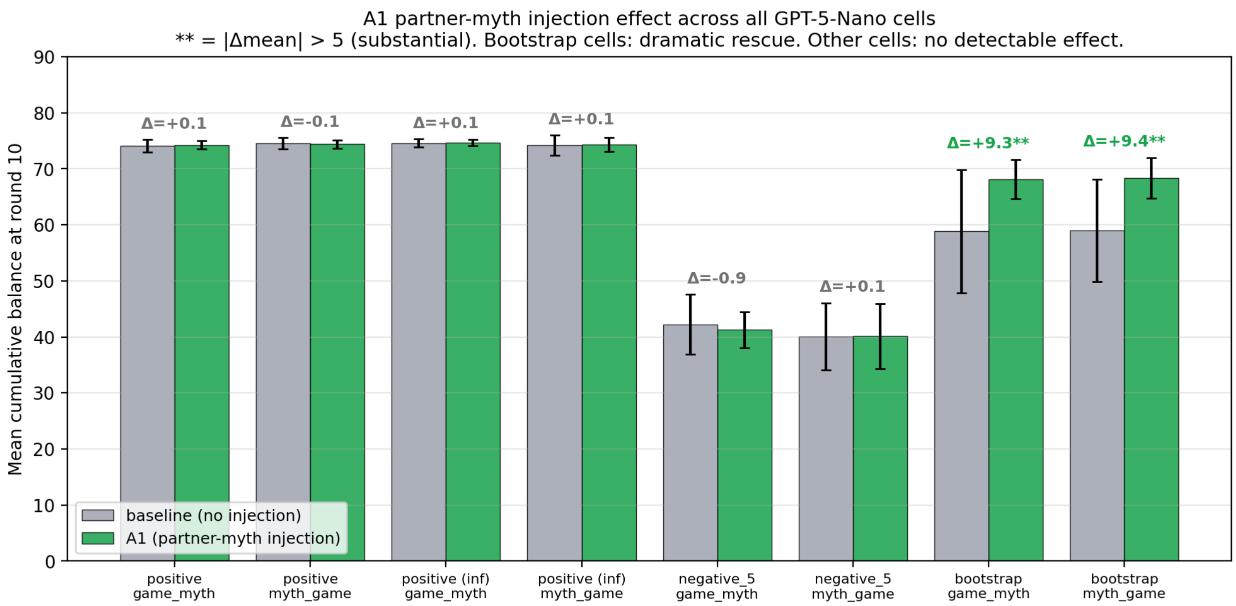

Figure 9: A1 (partner-myth injection) Delta-mean across all eight GPT-5-Nano noise cells; the rescue is concentrated in bootstrap cells and absent in positive and negative_5 cells.

6.6 A.6 Deferred analyses

The current draft omits three analyses that should be added before a stronger later-conference submission. First, embedding-based myth similarity should replace lexical Jaccard as the main semantic convergence measure. Second, a blinded qualitative codebook should classify recurring myth motifs and explicit reciprocity language. Third, lagged behavioural models should test whether myth features at round t predict send and return decisions at round $t+1$ after controlling for the previous game ledger.

6.7 A.7 Prompt templates

The exact prompts live in `config/experiments_noisy.yaml`. For a submission supplement, copy the system prompt, the four trust-game round prompts, and the two myth-writing prompts verbatim from that file, together with the exact experiment-set names used for the final manifest.